

3.	B	At $t=6$ s, $S_A - S_B = 54 \rightarrow (15 \times 6) - (\frac{1}{2} \times a_B \times 6^2) = 54 \rightarrow a_B = 2.0 \text{ m s}^{-2}$ Let the time for B to catch up with A be T. $S_A = S_B$ $15 \times T = \frac{1}{2} \times 2 \times T^2 \rightarrow T = 15.0 \text{ s}$ Hence additional time $= 15 - 6 = 9 \text{ s}$.
4.	D	The force on the rocket must be equal and opposite to the force on the ball